

DBMS –PAYMENTS SYSTEM MANAGEMENT SYSTEM

SAVANA–E-Commerce Order Management Database System

1.Database Design

1.1 Payment Table

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Related order ID
Customer_ID	Related customer ID
Payment_Date	Date of payment
Payment_Mode	UPI / Credit Card / Debit Card / Net Banking
Amount	Amount paid
Payment_Status	Successful / Pending / Failed

2.Table relationship

SAVANA_ORDERS → SAVANA_PAYMENT

Relationship: One-to-One / One-to-Many depending on payment design.

ORDER_ID in SAVANA_PAYMENT is related to ORDER_ID in SAVANA_ORDERS.

3. Table Creation

3.1 Create Payment Table

```
SQL> CREATE TABLE SAVANA_PAYMENT (  
2     PAYMENT_ID NUMBER(5) PRIMARY KEY,  
3     ORDER_ID NUMBER(5),  
4     CUSTOMER_ID NUMBER(5),  
5     PAYMENT_MODE VARCHAR2(30),  
6     PAYMENT_DATE DATE,  
7     AMOUNT NUMBER(10,2),  
8     PAYMENT_STATUS VARCHAR2(30)  
9 );
```

Table created.

4. Payment Data Insertion

```
SQL> INSERT INTO SAVANA_PAYMENT  
2 VALUES (501, 1001, 101, 'UPI',  
3 TO_DATE('21-09-2026','DD-MM-YYYY'),  
4 25998.00, 'Successful');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO SAVANA_PAYMENT  
2 VALUES (502, 1002, 102, 'Credit Card',  
3 TO_DATE('21-09-2026','DD-MM-YYYY'),  
4 1999.00, 'Successful');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO SAVANA_PAYMENT  
2 VALUES (503, 1003, 103, 'Debit Card',  
3 TO_DATE('20-09-2026','DD-MM-YYYY'),  
4 1598.00, 'Failed');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO SAVANA_PAYMENT  
2 VALUES (504, 1004, 101, 'Net Banking',  
3 TO_DATE('19-09-2026','DD-MM-YYYY'),  
4 1499.00, 'Pending');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO SAVANA_PAYMENT  
2 VALUES (505, 1005, 104, 'UPI',  
3 TO_DATE('18-09-2026','DD-MM-YYYY'),  
4 4999.00, 'Successful');  
  
1 row created.  
  
SQL>  
SQL> COMMIT;  
  
Commit complete.
```

5.Manage Payment Transaction

5.1 Display Successful Payment

```
SQL> SELECT * FROM SAVANA_PAYMENT;
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
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PAYMENT_STATUS

501 Successful	1001	UPI	21-SEP-26	25998
502 Successful	1002	Card	21-SEP-26	1999
503 Successful	1003	Cash	20-SEP-26	1598

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
------------	----------	--------------	-----------	----------------

PAYMENT_STATUS

504 Failed	1004	UPI	19-SEP-26	1499
505 Successful	1005	Net Banking	18-SEP-26	4999
1 Successful	1001	UPI	21-SEP-26	25998

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
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PAYMENT_STATUS

2 Successful	1002	Credit Card	21-SEP-26	1999
3 Successful	1003	Debit Card	20-SEP-26	1598
4 Pending	1004	UPI	19-SEP-26	1499

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
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PAYMENT_STATUS

5 Successful	1005	Cash on Delivery	18-SEP-26	4999
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10 rows selected.

6.Successful Transaction :

```
SQL> SELECT *
  2 FROM SAVANA_PAYMENT
  3 WHERE PAYMENT_STATUS = 'Successful';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
501	1001	UPI	21-SEP-26	25998
502	1002	Card	21-SEP-26	1999
503	1003	Cash	20-SEP-26	1598
505	1005	Net Banking	18-SEP-26	4999
1	1001	UPI	21-SEP-26	25998
2	1002	Credit Card	21-SEP-26	1999
3	1003	Debit Card	20-SEP-26	1598
5	1005	Cash on Delivery	18-SEP-26	4999

8 rows selected.

7. Failed Transaction :

```
SQL> SELECT *
2 FROM SAVANA_PAYMENT
3 WHERE PAYMENT_STATUS IN ('Failed', 'Pending');
```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	PAYMENT_MODE	PAYMENT_D
503	1003	103	Debit Card	20-SEP-26
1598	Failed			
504	1004	101	Net Banking	19-SEP-26
1499	Pending			

8.Modify Payment Status :

```
SQL> UPDATE SAVANA_PAYMENT
2 SET PAYMENT_STATUS = 'Successful'
3 WHERE PAYMENT_ID = 4;
```

1 row updated.

```
SQL>
```

```
SQL> COMMIT;
```

Commit complete.

9.ANALYZE PAYMENT METHODS :

9.1 COUNT TRANSACTIONS BY PAYMENT METHOD :

```
SQL> SELECT PAYMENT_MODE,
2 COUNT(*) AS TOTAL_TRANSACTIONS
3 FROM SAVANA_PAYMENT
4 GROUP BY PAYMENT_MODE;
```

PAYMENT_MODE	TOTAL_TRANSACTIONS
UPI	2
Credit Card	1
Debit Card	1
Net Banking	1

9.2TOTALAMOUNT BY PAYMENT METHOD :

```
SQL> SELECT PAYMENT_MODE,  
2          SUM(AMOUNT) AS TOTAL_AMOUNT  
3 FROM SAVANA_PAYMENT  
4 GROUP BY PAYMENT_MODE;
```

PAYMENT_MODE	TOTAL_AMOUNT
-----	-----
UPI	30997
Credit Card	1999
Debit Card	1598
Net Banking	1499

10. PAYMENT TRANSACTION REPORT :

```
SQL> SELECT
2     PAYMENT_ID,
3     ORDER_ID,
4     CUSTOMER_ID,
5     PAYMENT_MODE,
6     PAYMENT_DATE,
7     AMOUNT,
8     PAYMENT_STATUS
9 FROM SAVANA_PAYMENT
10 ORDER BY PAYMENT_DATE DESC;
```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	PAYMENT_MODE	PAYMENT_D
------------	----------	-------------	--------------	-----------

501	1001	101	UPI	21-SEP-26
25998	Successful			
502	1002	102	Credit Card	21-SEP-26
1999	Successful			
503	1003	103	Debit Card	20-SEP-26
1598	Successful			

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	PAYMENT_MODE	PAYMENT_D
------------	----------	-------------	--------------	-----------

504	1004	101	Net Banking	19-SEP-26
1499	Pending			
505	1005	104	UPI	18-SEP-26
4999	Successful			